



Research paper

Flow regime identification from acoustic sensors in pipe flow using deep neural networks

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ABSTRACT

In contemporary industrial processes, precise flow signal measurement is crucial for several reasons, such as enabling condition-based monitoring of pipeline health, preventing damage to expensive materials/equipment, minimizing environmental pollution, and ensuring the safety of operational personnel. In this work, a novel method for flow regime identification using advanced signal denoising techniques and deep neural networks is proposed. The time series data is collected from passive acoustic sensors that capture flow-generated acoustic signals on pipes within an experimental flow loop. The data is obtained under various flow conditions, with particular emphasis on slug flows, given their inherently high-noise nature. The denoising methods used here are an amalgamation of whitening procedures, autocorrelation filtering, bandpass filtering, and highly sophisticated deep neural networks. Notably, the first three methods are ingeniously adapted to facilitate the training of deep neural networks, eliminating the necessity for field data during the training process. Subsequently, the application of deep neural networks proves highly effective in refining signals associated with slug flows amidst extreme noise. This refinement allows for accurate determination of the slug regime type and its associated frequencies. Our approach to deep neural networks is particularly intriguing, as it leverages time-frequency segments for data denoising and incorporates artificial noise during training. This stands in contrast to existing neural network methodologies that predominantly focus on mapping inputs to outputs. Consequently, this technique introduces a cheap and fast method for flow lines monitoring where high levels of process noise are present, compared to existing extensive other approaches such as fiber optics monitoring or intrusive sensing. The experimental results demonstrate that the proposed technique benchmarking advanced methods, including the independent component analysis (ICA) and the eigenvalue decomposition (EGD) based approaches. In the presence of strong random noise, the presented methodology achieved an SNR of 30.2 dB, whereas the EGD and ICA methods attained SNR values of 15.3 dB and 17.5 dB, respectively, highlighting the superiority of the proposed approach.

1. Introduction

Gas–liquid flow regimes in production pipelines still present a great interest in terms of understanding the regimes associated and their consequences on the assets especially in unwanted flow regimes such as slug flows. There is a considerable research effort in both academia and the industry to monitor and control such flow regimes in order to optimize process engineering. Slug flows can be observed in multiple settings across a range of industries; in the petroleum industry in particular both dynamic slugging and terrain slugging exist in different parts of the production chain. In this aspect, monitoring is crucial

for the reliability of the pipeline transport system and the safety of the infrastructure. There are several techniques for pipeline or pipe flow monitoring starting from the basic pressure gauge application to the recent wide use of fiber optics (Kabiri-Samani and Borghei, 2010; Johannessen et al., 2012). Other sensing systems relying on either capacitive or resistive characteristics of the flow associated were used (Barnea et al., 1980). Non-intrusive flow sensing is a cost-effective approach, especially considering the advances in sensor technology. In current practices, fiber optics are used for distributed acoustic sensing

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(DAS) for flow line monitoring either in sub-sea risers, in-well tubulars, or land pipelines. However, the use of fiber optics is expensive and process intensive making this approach not an ideal solution for common flow monitoring such as flow regime changes in long pipelines. On the other hand, simple acoustic sensors, based on piezoelectric principles such as microphones were used for flow regime studies, however with simple approaches that cannot be used in real-life pipelines due to the presence of process noise for instance (Fossa et al., 2003; Gu and Gue, 2008; Elforjani and Mba, 2008; El-Alej et al., 2013).

Given the advancements in smart sensors, non-intrusive flow monitoring is a viable strategy. Such advanced flow sensing methods can be used for valve leak detection, flow condition checking, flow pattern recognition, and flow rate measurements, which are of paramount importance for fluid flow monitoring in facilities both at the surface level and down-hole (Hansen et al., 2019). In most cases, the complexity of these operations is due to the simultaneous presence of gas and liquid in the flow lines, making conventional flow sensing techniques unreliable. The volume of gas and liquid in the fluid introduces several interfacial conditions due to the two phases producing different flow patterns, which control the way the gas and liquid are distributed. This behavior significantly affects the method for detecting a potential leak from a valve in the flow path or the total flow in a straight pipe. Several reports have been published identifying the nature of the gas–liquid interactions and the conditions used to classify the flow regimes observed, such as slug flow (Sarkodie et al., 2019; Al-Kayiem et al., 2017; Shaban and Tavoularis, 2016). In this regard, one or more passive acoustic sensors placed in a flow line can be utilized to obtain valuable information for checking the condition of valves, the measurement of the flow rate of individual components, and the classification of flow regimes.

The literature details a vast array of techniques for measuring slug flow, such as electrical impedance, radiation attenuation techniques, ultrasonic methods, and optical and video methods (Do Nascimento Arrais et al., 2024; Jin et al., 2024; Gonçalves and Mazza, 2022; Guo et al., 2023). Another effective method for monitoring two-phase or multi-phase flow, commonly used in industry, is electrical resistance tomography (ERT). In Ciani et al. (2024), the authors integrated a dual-plane ERT system with the cross-correlation technique in order to measure a two-phase gas–liquid flow. The study in Hanus et al. (2017) introduced three primary flow regimes—plug, bubble, and transitional plug-bubble—and utilized signal features in both the time and frequency domains to develop an artificial neural network for recognizing two-phase flow in a horizontal pipeline. Additionally, principal component analysis (PCA) was applied to reduce the number of features required (Ju et al., 2022). Although the literature covers using various methods for two-phase flow measurements, it appears that the detrimental effects of noise on sensor flow measurements have not been adequately addressed, leading to potential inaccuracies in determining two-phase flow rates.

Most of the acoustic signals collected from pipe flows, either in a laboratory environment or an engineering process, are subject to additional noise that could have different origins related to the accessories associated with the pipe flow or due to the environment surrounding the flow system. Therefore, polishing the signals collected from acoustic sources becomes an important aspect of the monitoring system to ensure the capture of the flow characteristics targeted without jeopardizing the loss of crucial information from the flow regimes associated with these operations. The denoising techniques currently used are out-of-date for successful monitoring of the flow and to provide real-time flow checking.

The following are the key contributions:

- In this work, various denoising techniques are utilized to achieve the desired goals of identifying the flow regime. In order to accomplish this task, a flow pipe assembly with horizontal and

vertical sections is used and two sensors are placed in horizontal and vertical locations. The sensor data acquired at the vertical section is relatively clean (less noisy), whereas the sensor data acquired at the horizontal section requires extensive hovering. This confirms the severity of horizontal pipe monitoring (Mohammed et al., 2021).

- Several methods including bandpass filtering (Shenoi, 2005), wavelet denoising/thresholding (Antoniadis, 2007), whitening procedure (Iqbal et al., 2018; Liu et al., 2017), autocorrelation filtering (Liu et al., 2017) are used to generate the training data. Deep neural network is then used to enhance the signal of interest in order to identify the slug flows using acoustic sensors. The combination of the methods like whitening, autocorrelation, and wavelet-based filtering are used for the first time for the considered scenario.
- A method of denoising using a transform-domain deep neural network is used, in which one sample of the acquired data acts as a training point, hence reducing the requirement of large data sets for training.

The remainder of this paper is structured as follows. The experimental setup is covered in Section 2. Section 3 presents the proposed denoising approaches, followed by the presentation and discussion of experimental results in Section 4. Finally, Section 5 provides the conclusions drawn from the findings.

2. Experimental setup

The experimental work was conducted in a dedicated flow loop in which a mixture of gas and liquid can both flow reliably. The flow loop was closed and comprised of a feed tank that is connected to a pump operated with a variable speed drive (VSD). The pump with a maximum capacity of 100 L/min is connected to an electromagnetic flow meter. The flow meter is connected directly to the flow pipe made of transparent PVC of 3 inches diameter that has two long sections, a horizontal test section (HTS) of 3 m, followed by a vertical test section (VTS) of 2 m. The whole assembly with horizontal and vertical sections is shown in Fig. 1. The pipe flow returns back to the feed tank. The gas utilized in the experiments is compressed air supplied by a central compression system in the building, with a delivery pressure of 10 Bars. It was injected just downstream of the flow meter, and a check valve was installed to prevent backflow at the meter junction. Although the flow section is short in diameter and length, it was designed to rapidly generate slug flows that could be analyzed during the experiments. A visualization of the slug movement in horizontal and vertical sections is depicted in Figs. 2 and 3, respectively, for the gas flow rate Q_g of 32 L/min and liquid flow rate Q_l of 50 L/min. The room temperature was constantly controlled. The figures confirm the difficulty in measuring slug flow in the horizontal section as compared to the vertical section. Acoustic sensors (microphone Knowles Mic. Cond. Analog frequency range 280–5000 Hz) were affixed to the pipe walls using adhesive to facilitate proper acoustic coupling between the sensor and the pipe material.

A multi-channel acquisition system operated with LabVIEW from National Instruments was employed to capture signals from the acoustic sensors at various acquisition rates. The signals were collected for various flow regimes, generated at a range of superficial liquid and gas velocities, which were controlled using the VSD. The experiments covered a wide envelope of superficial liquid and gas velocities in order to assess the sensitivity of the acoustic system used as well as capture various noises present in the pipe flow. Samples of the signals collected in both horizontal and vertical sections are shown in Fig. 4. In the horizontal, section as expected from the slug regimes generated, the slug frequencies were longer than in the vertical section which was clearly seen in the signals collected from the vertical section. The ones collected from the sensors in the horizontal section were very noisy and

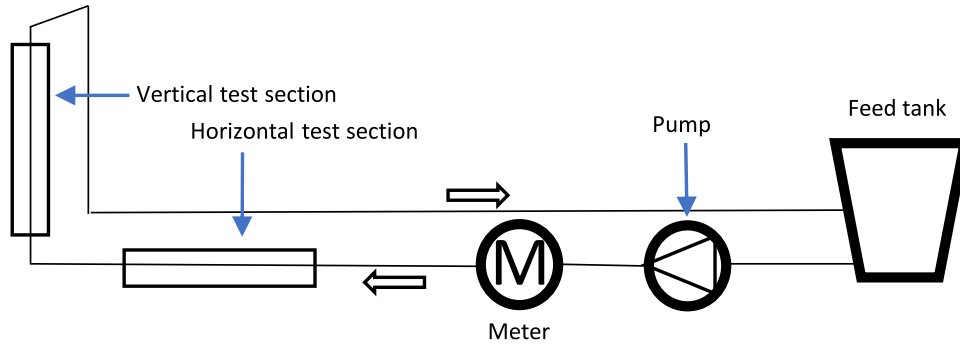


Fig. 1. Experimental setup schematics.

(a)



(b)



Fig. 2. Visualization of the liquid slug movement in horizontal section, (a) image shows the end of the liquid slug (slug tail) and (b) image shows the region where gas and liquid are stratified, the interface being wavy before the next slug arrives for volume flow rates of $Q_g = 32$ L/min and $Q_l = 50$ L/min.

could not exhibit clearly the slugs boundaries compared to the signals from the vertical section. Once the signals were collected over a period of minimum 3 min, there were stored and later analyzed on PC using denoising techniques that will be discussed in the next section.

3. Denoising techniques

Traditional denoising methods often rely on statistical techniques and signal processing algorithms that work effectively under certain conditions, particularly when the signal-to-noise ratio (SNR) is high. In such scenarios, where noise levels are low relative to the signal,

these methods can accurately recover the original signal from the noisy observations. Hence various traditional denoising methods including bandpass filtering (Shenoi, 2005), wavelet denoising/thresholding (Antoniadis, 2007), whitening procedure (Iqbal et al., 2018; Liu et al., 2017), and autocorrelation filtering (Liu et al., 2017) are used in this work. Using a method from the aforementioned methods gives a marginal improvement in VTS sensor data which is already relatively clean. However, no improvement was observed for the HTS sensor data due to the strong noise. In this work, a multi-step denoising method is proposed as follows:

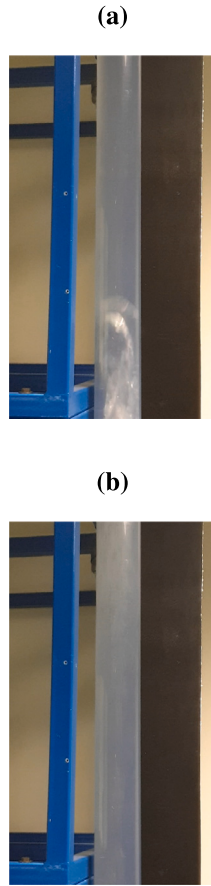


Fig. 3. Visualization of the liquid slug movement in vertical section, (a) image shows the end of the liquid slug (slug tail) and (b) image shows the region where gas and liquid are stratified, the interface has air bubbles before the next slug arrives for $Q_g = 32$ L/min and $Q_l = 50$ L/min.

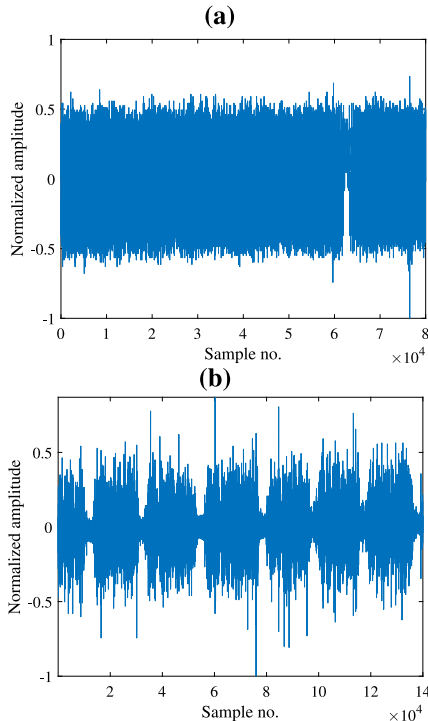


Fig. 4. Samples of the signal collected for the same flow regime at both, (a) horizontal and (b) vertical sections for volume flow rates of $Q_g = 32$ L/min and $Q_l = 50$ L/min.

- The VTS sensor data is refined further using the whitening procedure, autocorrelation filtering, wavelet denoising, and bandpass filtering in sequence.
- The honed data obtained in the previous step is corrupted using various types of noises.
- After processing the VTS sensor data, the cleaned and synthetically corrupted data are used for training the deep neural network.
- Finally, the HTS sensor data is polished using the trained deep neural network.

In the ensuing, the background of each of the denoising techniques (Shenoi, 2005; Antoniadis, 2007; Iqbal et al., 2018; Liu et al., 2017; Iqbal, 2023) is detailed to ensure self-containment and readability.

3.1. Whitening procedure and autocorrelation-based filtering

Since the autocorrelation-based filtering depends on stacked autocorrelations, therefore, the data from the sensor is divided into N channels each of length M samples. Each channel contains noisy measurements. The n th sample of the observation is given as

$$x_i(n) = s_i(n) + \eta_i(n), \quad i = 1, \dots, N \quad (1)$$

where s_i represents the signal under consideration and η_i is the noise component. The value of N is taken as 200. The nature of the noise is unknown and, therefore, assumed to be colored noise having power spectral density that is not flat like white noise. The fundamental presumption underlying the autocorrelation-based enhancement method is the additive white noise. However, when it comes to the environment, weather, or human activities, noise is commonly colored in practice. As a result, the noise autocorrelation may not be an impulse function. Consequently, the frequency response of the noise cannot be removed by altering the stacking autocorrelation value solely at the zero lag location. Here, a noise whitening approach that is based on linear prediction theory is used to generalize the method for colored noise. Since noise and the signal are not separable, the whole data is whitened. This does affect the results as the aim is to find the slug frequency instead of the exact shape of the signal s_i .

Let $x(n)$ represent the stationary colored noise. A wide-sense stationary random process can be approximated by an autoregressive (AR) process if the model's order is chosen to be sufficiently large (Haykin, 2002). Then, using L prior samples, a linear prediction filter with coefficients $\{c_k\}$ can be used to forecast the following sample as

$$\hat{x}[n] = \sum_{k=1}^L c_k x[n-k], \quad (2)$$

L denotes the prediction filter order and the prediction error can be expressed as follows:

$$e[n] = x[n] - \hat{x}[n]. \quad (3)$$

In order to have the optimal linear predictive coding (LPC) coefficients, we minimize the prediction error power of the L th order linear predictor. This process is equivalent to maximizing spectral flatness at the prediction error filter's output. For an AR(L) process, the optimal prediction coefficients are the AR(L) parameters. Thus, the prediction error filter performs the whitening operation and decorrelates the input AR process by generating a white signal at the output. Determining L is the model order selection problem that can be solved by using the fact that for a linear predictor of order p , the error power is constant, where L is the true order of the AR process and $p \geq L$ for the output to be white. In this study, p is taken as 100.

The whitening filter $F(z)$ in the z -transform domain is represented as follows:

$$F(z) = 1 - A(z), \quad (4)$$

and the linear prediction filter $A(z)$ is represented as

$$A(z) = \sum_{k=1}^L c_k z^{-k}.$$

The autocorrelation method of AR modeling can be used to calculate the prediction coefficients c_k . This entails predicting the autocorrelation sequence from the time series and using the Levinson–Durbin recursion to solve the Yule–Walker equations. The parameters of the whitening filter are calculated for each channel and then the noisy signal is whitened before using the denoising filter shown next.

The following steps are used to implement the autocorrelation-based strategy for denoising:

- Compute and stack the autocorrelation functions of each channel, i.e.,

$$a_s[\gamma] = \frac{1}{N} \sum_{i=1}^N (x_i \star x_i)[\gamma], \quad (5)$$

where $\star$ denotes the autocorrelation and $\gamma = -P+1, \dots, 0, \dots, P-1$ is the lag index of the autocorrelation function.

- Modify the autocorrelation function as

$$\hat{a}_s[\gamma] = \begin{cases} \frac{1}{2} (a_s[-1] + a_s[1]) & \text{if } \gamma = 0 \\ a_s[\gamma] & \text{otherwise} \end{cases} \quad (6)$$

The zero-lag value of the autocorrelation function is replaced with the neighboring values' average, $a_s[1]$ and $a_s[-1]$. The motivation behind this step is that additive white noise has an autocorrelation function $P\sigma^2\delta[\gamma]$, where $\delta[\gamma]$ is the discrete Dirac delta impulse function that gives non-zero value only at $\gamma = 0$, and σ^2 is the variance of noise after whitening of observation. As a result, the autocorrelation function $a_s[\tau]$ has a large peak at $\tau = 0$ that must be reduced by $P\sigma^2$; the average of the neighboring values yields an estimate of the correct zero-lag value of the signal-only autocorrelation function.

- Define the impulse response of the filter as a windowed version of the modified autocorrelation function, $h[\gamma] = \hat{a}_s[\gamma]w_d[\gamma]$, where

$$w_d[\gamma] = \begin{cases} 1 - |\gamma|/d & \text{if } |\gamma| \leq d \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

The truncation window is characterized by a length of $2d+1$. The truncation window $w_d[\gamma]$, when applied to the zero-lag region, filters out insignificant values in the filter, particularly those far from $\tau = 0$. Additionally, using an appropriate truncation window to shorten the filter length can enhance computational efficiency. There are a variety of truncation windows available (Proakis and Manolakis, 2006), however, in this work a simple triangle window is adopted.

- Convolve $h[\gamma]$ with the noisy signal of each channel. The resultant convolutions provide the N denoised signals, i.e.,

$$\hat{x}_i[n] = (h \star x_i)[n] \text{ for } i = 1, 2, \dots, N. \quad (8)$$

3.2. Wavelet-based denoising and bandpass filtering

The technique of bandpass filtering is widely used for improving the SNR. By rejecting the frequencies outside the signal's frequency range enhance the SNR. However, if the signal and noise have the same frequency band, this will cause signal attenuation and/or noise not to be completely removed. After the observation signal is whitened and autocorrelation filter is applied then it is bandpass filtered from 10 Hz to 500 Hz. This range is selected in order to filter out any low-frequency components. Finally, wavelet thresholding is used to get a clearer copy of the signal. Wavelets are used to extract the high and low-frequency components in this method. It is assumed that the high-frequency components represent noise. Of course, this necessitates the

use of a threshold and a basis function (mother wavelet). In this work, a 14-level wavelet decomposition is used with the basis function of Daubechies (*db15*) and soft minimax thresholding (Daubechies, 1992). For more details on this matter see the Matlab wavelet toolbox (Misiti et al., 2021).

3.3. Deep neural networks based denoising

The methods listed above work well with the VTS sensor data that has a relatively high SNR. The results, however, were not promising when applied to the data from the HTS sensor. Hence, a deep neural network-based signal enhancement (noise suppression) approach, DeepSeg, is used and adapted for the presented scenario. Details about this method are provided in the following.

3.3.1. Training data set

One of the main requirements of deep neural networks is the training data set (Han et al., 2021; Iqbal, 2023). The VTS sensor data is used for this purpose. The cleaned data is first corrupted using various color noise sequences.

For this purpose, the fast fractional differences algorithm (Xu, 2019) is utilized for the noise generation. Two parameters are required for this noise generation approach: fractional memory parameter d (half of the spectral index) and degree of correlation ϕ ($= 0.99$). To generate various types of noise sequences, the value of d is randomly chosen from the range $[-2, 2]$. The noise realization for $d = -2$ has a high correlation among samples and low frequency content as contrast to $d = 2$, whereas $d = 0$ means no correlation and flat spectrum (white noise).

After adding various forms of noise to the VTS sensor data, the target is to design a deep neural network that produces clean data when given the synthetically corrupted data as input. For the development of a deep neural network, the following approach is adopted.

3.3.2. Preprocessing

To begin, the one-dimensional sensor data is converted to two-dimensional data. This is possible, thanks to the short-time Fourier transform. However, it involves complex numbers, whereas neural networks work well with real numbers and give the best prediction accuracy. Furthermore, it is not possible to have the last fully connected layer in the proposed deep neural network design that is discussed in the next section. As a consequence, it provides more adaptability in network design when using only real numbers. For this purpose, a discrete cosine transform is used and termed short-time discrete cosine transform (STDCT). In STDCT, the discrete cosine transforms are applied to segmented portions of the data, which are created by sliding a window across the entire data sequence. In this work, a segment refers to a window section to which DCT is applied. The DCT type-IV (Britanak and Rao, 2018) used here is defined for a window size of W as:

$$y(k) = \sqrt{\frac{2}{W}} \sum_{n=1}^W x(n) \cos\left(\frac{\pi}{4W} (2n-1)(2k-1)\right). \quad (9)$$

For finding the inverse, n and k are interchanged in the above definition of DCT. The STDCT is used to convert the data to a 2D time–frequency representation using a rectangular window of size $W = 128$ samples and a 90% overlap. Hence, time–frequency domain spectra of the data are obtained and a deep neural network is used to extract the clean spectra from the noisy spectra. Finally, a method akin to the inverse short-time Fourier transform is employed (Sharpe, 2023) is used to obtain the time-domain data.

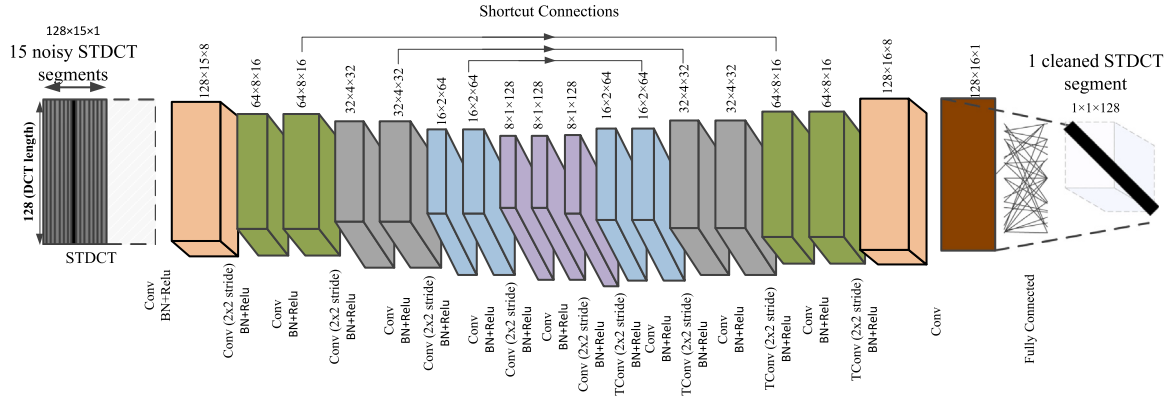


Fig. 5. SNR enhancement deep convolutional neural network architecture. The 15 segments of the noise-contaminated signal's time–frequency representation are fed to the network., whereas the prediction of the 1 segment is taken as output. The dimension of each layer is indicated above it, and the information is presented in the format “frequency bins $\times$ time samples $\times$ channels”. The 2D input data is first downsampled by passing it through convolutional layers (Conv) of size 3×3 and stride 2×2 . This is followed by upsampling using transpose convolutional layers (TConv). The kernel size of the last convolutional layer is 1×1 whereas the network concludes with a fully connected layer. Batch normalization (BN) and shortcut connections are used to improve convergence during the training phase.

3.3.3. Deep neural network

In this study, the denoising problem is framed as a supervised deep learning task, where a neural network learns the sparse representation of the signal using training data set that includes ground truth information.

The sensor's data, $\mathcal{Y}$, is represented in the time–frequency domain as a superposition of the signal of interest, $\mathcal{X}$, and additive noise, $\mathcal{N}$, that is,

$$\mathcal{Y} = \mathcal{X} + \mathcal{N}. \quad (10)$$

Given a noisy spectra segment $\{\mathcal{Y}_n\}_{n=1}^T$ and clean spectra segment $\{\mathcal{X}_n\}_{n=1}^T$, our goal is to model the mapping function f to estimate ‘denoised’ spectra segment $\{f(\mathcal{Y}_n)\}_{n=1}^T$ by minimizing the ℓ_2 norm, that is,

$$\min \sum_{n=1}^T \|\mathcal{X}_n - f(\mathcal{Y}_n)\|_2^2. \quad (11)$$

Particularly, a deep neural network is used to learn the mapping function f . The past $\lfloor \Delta/2 \rfloor$ noisy spectra segments $\{\mathcal{Y}_i\}_{i=n-\lfloor \Delta/2 \rfloor}^n$, the future $\lfloor \Delta/2 \rfloor$ noisy spectra segments $\{\mathcal{Y}_i\}_{i=n+1}^{n+\lfloor \Delta/2 \rfloor}$ and the current noisy spectra segment $\mathcal{Y}_n$ are considered to denoise the current spectra segment $\mathcal{Y}_n$, that is,

$$\sum_{n=1}^T \|\mathcal{X}_n - f(\mathcal{Y}_{n-\lfloor \Delta/2 \rfloor}, \dots, \mathcal{Y}_{n+\lfloor \Delta/2 \rfloor})\|_2^2. \quad (12)$$

The value of segments used is set as $\Delta = 15$, which is determined using a sensitivity analysis procedure. This strategy to neural network design for noise removal is analogous to the filter design approach in the digital signal processing field (Proakis and Manolakis, 2006), and it is more rational than merely mapping input to output with a neural network for performing the denoising task.

In this study, a deep neural network is structured as a series of convolutional layers with descending followed by ascending sizes (similar to U-net architecture (Ronneberger et al., 2015)). This is inspired by the autoencoders which have the potential to learn sparse features of the data in order to achieve an optimization objective.

The deep convolutional neural network that is used in this study is depicted in Fig. 5 and described as follows. The input to the network is 15 noisy STDCT segments, with the cleaned middle segment as the output. The network setup consists of convolutional layers which are followed by transposed convolutional layers. As can be seen from the figure the convolutional and fully connected layers are the network's final two layers, respectively. To make sure that the final layer's size corresponds to the intended output size, a fully connected layer is

added. The convolutional and transpose convolutional layers have filter size of 3×3 each, whereas the number of filters ranges from 8 to 128. The network's second last layer, i.e., the final convolutional layer has one filter with a size of 1×1 . Except for the final convolutional and fully connected layer, the rectified linear unit (ReLU) layer and the batch normalization (BN) layer follow each convolutional and transpose convolutional layer. The final fully connected layer of the network has 128 neurons.

The following parameters are used for convolutional layers and transposed convolutional layers: The stride value is adjusted to either 1 or 2, based on whether resampling is needed; this includes downsampling for convolutional layers and upsampling for transpose convolutional layers. Convolutional layers utilize padding to make the output the same size as the input when stride equals 1, whereas the output size is $\lceil [In_{Size}/St] \rceil$ (downsampled) when the stride is greater than 1, where In_{Size} is the input's height or width, and St is the stride in that dimension. On the other hand, transpose convolutional layers use cropping (reduction in output size) to make the output equal to $In_{Size} * St$ (upsampled). The same amount of padding is added to the top and bottom, as well as to the left and right. However, when the value of vertical padding required is an odd number, extra padding is added to the bottom. Likewise, if the value of horizontal padding needed is odd, additional padding is added to the right side.

In order to enhance the prediction performance and training convergence shortcut connections are employed in this study, as suggested by He et al. (2016). When shortcut connections are used, the signal feeding a layer is also added to the output of a layer higher up the stack.

4. Results and discussion

The present work has revealed that using simple acoustic sensors that are widely available in the market could be a robust method for flow line monitoring with the right signal analysis and denoising approaches. The results obtained by using the aforementioned methods are confirmed visually by the videos taken. This study demonstrated the ability of advanced denoising techniques in polishing collected signals from process noise and surrounding noise without losing the information from the flow regime investigated. The physical sensing setup used is simple and does not require any special care, meaning it can be ported to a real pipeline system for monitoring, provided that the right signal acquisition and treatment means are present. Using the setup described in Section 2, collected signals in this study were cleaned with the algorithms described in the above sections. Particularly, the current investigation was focused on characterizing

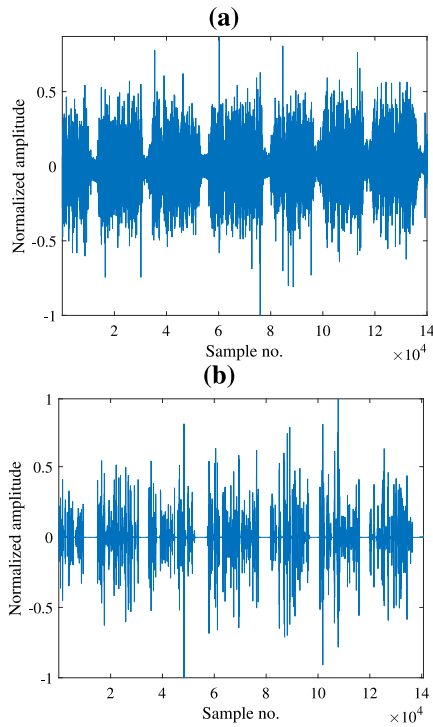


Fig. 6. (a) Raw signal collected from the VTS acoustic sensor, (b) denoised signal obtained from (a) for volume flow rates of $Q_g = 32$ L/min and $Q_l = 50$ L/min.

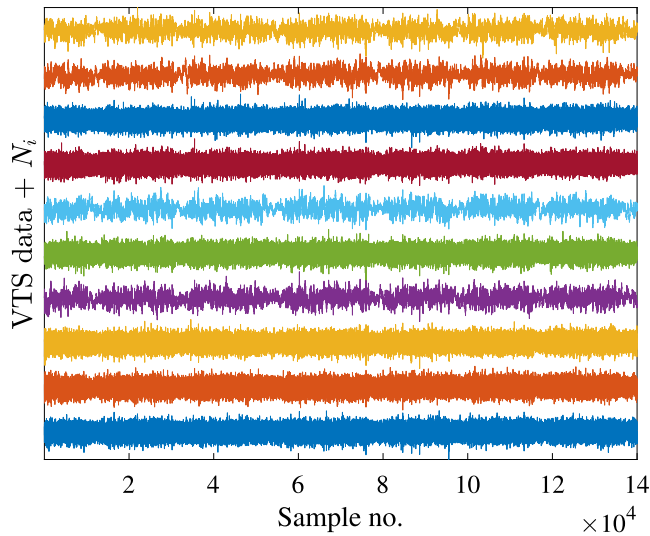


Fig. 7. Copies of VTS sensor data with various added noise (N_i) in order to make SNR = 0 db.

the slug flow regime due to its noisy nature from both the dynamic flow behavior and the environment. The results have shown that the slug flow regimes observed can be clearly identified and qualified. The slug frequencies and lengths from the collected signals could be quantified and precisely measured depicting the periodic or aperiodic nature of the slugging flow.

The novelty of the present work relies on the fact that the VTS sensor data are denoised using the techniques mentioned in Sections 3.1 and 3.2. The results of this processing are shown in Fig. 6. It can be seen from the figure that the methodology works well to denoise the VTS sensor data. Next, the denoised VTS sensor data is corrupted by various types of noise and some samples are depicted in Fig. 7. This data is then

Table 1

Comparison of various recovery methods.

Recovery method	SNR (dB)
EGD	15.3
ICA	17.5
Proposed method	30.2

used in deep neural network training explained in Section 3.3. The VTS sensor data with synthetically added noise were used as the input to the deep neural network, whereas the output was the cleaned VTS sensor data.

The training curve for the deep neural network is depicted in Fig. 8. It shows that the training took around 11 epochs to finish on GPU. The following parameters are used for optimization: The weights of all the layers are initialized as in Glorot and Bengio (2010), and the biases are set to zero. The network is trained with Adam (Kingma and Ba, 2017) and a 64 mini-batch size using backpropagation with gradient descent optimization. The learning rate is started from $lr = 10^{-2}$ with $\epsilon = 10^{-8}$, $\beta_1 = 0.9$, and $\beta_2 = 0.999$ and then it is decreased by a factor of 0.9 every epoch. When the validation loss does not decrease after 5 epochs, the training is terminated. The gradients are calculated using the following loss function:

$$\mathbb{E} = \sum_{q=1}^{128} \frac{1}{2} [t(q) - p(q)]^2. \quad (13)$$

where $p()$ denotes the network predictor and $t()$ denotes the target output. To reduce the problem of over-fitting, a regularization term is added to the loss function $\mathbb{E}$ (Murphy, 2012), as follows:

$$\mathbb{E}_r = \mathbb{E} + \frac{\lambda}{2} \mathbf{w} \mathbf{w}^T. \quad (14)$$

The regularization factor λ is usually small and in this work is chosen to be 0.00005.

After training the deep neural network, the HTS data shown in Fig. 9a was fed to the network. The raw signal and a polished signal after the treatment using deep neural network is shown in Fig. 9a and Fig. 9b, respectively. The output is a high-quality version of the data. Tremendous improvement in SNR is being observed using the already trained deep convolutional neural network. The slug flow can easily be identified in the cleaned copy of the data.

These findings are substantial since it is very hard experimentally to qualify slugging flow regimes in a clear fashion. Commonly pressure transducers are used to qualify the slug frequencies and periodicity, however, pressure transducers lack the sensitivity and the time response needed. On the other hand, acoustic transducers can effectively track slugs in the real time, and respond to any flow alteration resulting from the change in the pressure field inside the pipe flow that is translated into an acoustic field. The present study demonstrates that this approach is both robust and straightforward to implement, provided that appropriate acquisition methods and data analysis tools are used. The deep learning analysis used in this study provided improved processing of the collected signals and demonstrated that characteristic signals of slug flow regimes can be recovered with minimal noise—something that is challenging to achieve with conventional methods.

Table 1 presents a comparison of recovery performance between the proposed deep learning method and various state-of-the-art techniques, including generalized eigenvalue decomposition (EGD) (Zhou and Li, 2016) and independent component analysis (ICA) (Sahonero-Alvarez and Calderon, 2017). All methods are tested under the presence of white Gaussian noise that is added to the data presented in Fig. 6b. This is to ensure that the groundtruth is available for qualitative analysis. The SNR of the noisy signal is set to be 0dB. It is important to highlight that the ICA method used here is the Joint Approximation Diagonalization of Eigenmatrices (JADE) approach. Remarkably, the proposed

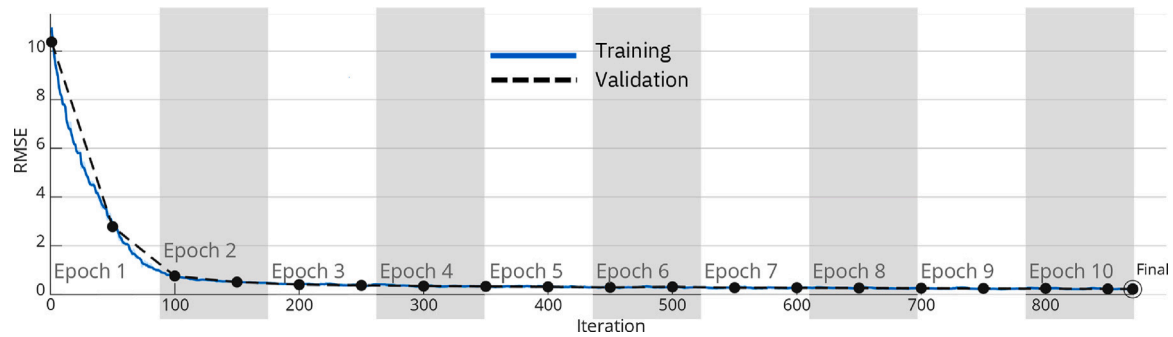


Fig. 8. The convergence curve root mean square error (RMSE) versus iterations for the deep neural network training.

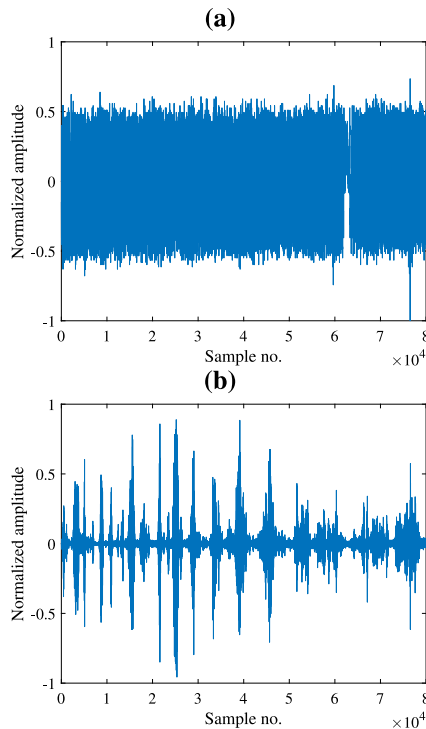


Fig. 9. (a) Raw signal collected from the HTS acoustic sensor, (b) denoised signal using proposed deep learning method obtained from (a) for volume flow rates of $Q_g = 32$ L/min and $Q_l = 50$ L/min.

deep learning method shows better signal recovery performance than these leading techniques.

The signals shown in the present study are examples of real data that can be collected from a pipeline with gas and liquid flowing and where slug flow regimes can be present. The acoustic signals and the methods presented here can serve as the basis of more sophisticated systems that could be implemented in a pipeline monitoring station for flow assurance and pipeline integrity. Instead of relying on kilometers of fiber optics, as is common in current practices, a few strategically placed piezo-acoustic sensors can be used to monitor flow in metal pipelines. These sensors can identify the flow regime, register slug frequency, and enable pipeline operators to take appropriate actions when flow regime changes are detected, thereby supporting operational integrity.

5. Conclusions

The present study showed that simple acoustic sensors could be used in any environment even in the presence of high background noise

for flow monitoring and characterization. Slug flows were used as an example in the experiments shared here and using advanced signal denoising techniques discussed in this paper showed the ability to clearly identify the flow features and parameters from noisy raw acoustic signals. The present method could play a game-changing role for flow monitoring in process engineering where wide bands of noise are expected eliminating the need for using high-end physical sensing tools. Moreover, the numerical results clearly indicate that the proposed algorithm offers significant improvements over existing advanced methods, including the EGD-based and ICA-based approaches. The superior SNR performance of 30.2 dB achieved by the proposed method, compared to the 15.3 dB and 17.5 dB achieved by the EGD and ICA methods, respectively, underscores the effectiveness and robustness of the proposed approach in handling noise. These findings validate the proposed algorithm as a more reliable and efficient solution for the problem at hand. This study opens the door for additional developments in terms of packaging and real-time monitoring.

CRedit authorship contribution statement

Naveed Iqbal: Formal analysis, Data curation, Conceptualization. **Azzedine Zerguine:** Methodology, Investigation, Conceptualization. **Mohamed Nabil Noui-Mehidi:** Project administration, Investigation, Funding acquisition, Formal analysis. **Mohamed Larbi Zeghlache:** Resources, Project administration, Methodology, Investigation. **Abdulmajid Lawal:** Formal analysis, Data curation, Conceptualization. **Ali Al-Shaikh:** Resources, Methodology, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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